

can sometimes display “convergence” that is only apparent, not real. The remedy is to try two different transformations as a check.

Since convergence acceleration is so much more difficult for a series of positive terms than for an alternating series, occasionally it is useful to convert a series of positive terms into an alternating series. Van Wijngaarden has given a transformation for accomplishing this [6]:

$$\sum_{r=1}^{\infty} v_r = \sum_{r=1}^{\infty} (-1)^{r-1} w_r \quad (5.3.20)$$

where

$$w_r \equiv v_r + 2v_{2r} + 4v_{4r} + 8v_{8r} + \cdots \quad (5.3.21)$$

Equations (5.3.20) and (5.3.21) replace a simple sum by a two-dimensional sum, each term in (5.3.20) being itself an infinite sum (5.3.21). This may seem a strange way to save on work! Since, however, the indices in (5.3.21) increase tremendously rapidly, as powers of 2, it often requires only a few terms to converge (5.3.21) to extraordinary accuracy. You do, however, need to be able to compute the v_r ’s efficiently for “random” values r . The standard “updating” tricks for sequential r ’s, mentioned above following equation (5.3.1), can’t be used.

Once you’ve generated the alternating series by Van Wijngaarden’s transformation, the Levin d transformation is particularly effective at summing the series [8]. This strategy is most useful for linearly convergent series with ρ close to 1. For logarithmically convergent series, even the transformed series (5.3.21) is often too slowly convergent to be useful numerically.

As an example of how to call the routines `Epsalg` or `Levin`, consider the problem of evaluating the integral

$$I = \int_0^{\infty} \frac{x}{1+x^2} J_0(x) dx = K_0(1) = 0.4210244382 \dots \quad (5.3.22)$$

Standard quadrature methods such as `qromo` fail because the integrand has a long oscillatory tail, giving alternating positive and negative contributions that tend to cancel. A good way of evaluating such an integral is to split it into a sum of integrals between successive zeros of $J_0(x)$:

$$I = \int_0^{\infty} f(x) dx = \sum_{j=0}^{\infty} I_j \quad (5.3.23)$$

where

$$I_j = \int_{x_{j-1}}^{x_j} f(x) dx, \quad f(x_j) = 0, \quad j = 0, 1, \dots \quad (5.3.24)$$

We take x_{-1} equal to the lower limit of the integral, zero in this example. The idea is to evaluate the relatively simple integrals I_j by `qromb` or Gaussian quadrature, and then accelerate the convergence of the series (5.3.23), since we expect the contributions to alternate in sign. For the example (5.3.22), we don’t even need accurate values of the zeros of $J_0(x)$. It is good enough to take $x_j = (j+1)\pi$, which is asymptotically correct. Here is the code:

```

levex.h  Doub func(const Doub x)
        Integrand for (5.3.22).
        {
            if (x == 0.0)
                return 0.0;
            else {
                Bessel bess;
                return x*bess.jnu(0.0,x)/(1.0+x*x);
            }
        }

Int main_levex(void)
This sample program shows how to use the Levin  $u$  transformation to evaluate an oscillatory
integral, equation (5.3.22).
{
    const Doub PI=3.141592653589793;
    Int nterm=12;
    Doub beta=1.0,a=0.0,b=0.0,sum=0.0;
    Levin series(100,0.0);
    cout << setw(5) << "N" << setw(19) << "Sum (direct)" << setw(21)
        << "Sum (Levin)" << endl;
    for (Int n=0; n<=nterm; n++) {
        b+=PI;
        Doub s=qromb(func,a,b,1.e-8);
        a=b;
        sum+=s;
        Doub omega=(beta+n)*s;      Use  $u$  transformation.
        Doub ans=series.next(sum,omega,beta);
        cout << setw(5) << n << fixed << setprecision(14) << setw(21)
            << sum << setw(21) << ans << endl;
    }
    return 0;
}

```

Setting eps to 1×10^{-8} in `qromb`, we get 9 significant digits with about 200 function evaluations by $n = 8$. Replacing `qromb` with a Gaussian quadrature routine cuts the number of function evaluations in half. Note that $n = 8$ corresponds to an upper limit in the integral of 9π , where the amplitude of the integrand is still of order 10^{-2} . This shows the remarkable power of convergence acceleration. (For more on oscillatory integrals, see §13.9.)

CITED REFERENCES AND FURTHER READING:

- Weniger, E.J. 1989, "Nonlinear Sequence Transformations for the Acceleration of Convergence and the Summation of Divergent Series," *Computer Physics Reports*, vol. 10, pp. 189–371.[1]
- Abramowitz, M., and Stegun, I.A. 1964, *Handbook of Mathematical Functions* (Washington: National Bureau of Standards); reprinted 1968 (New York: Dover); online at <http://www.nrl.com/aands>, §3.6.[2]
- Mathews, J., and Walker, R.L. 1970, *Mathematical Methods of Physics*, 2nd ed. (Reading, MA: W.A. Benjamin/Addison-Wesley), §2.3.[3]
- Numerical Recipes Software 2007, "Derivation of the Levin Transformation," *Numerical Recipes Webnote No. 6*, at <http://www.nr.com/webnotes?6> [4]
- Smith, D.A., and Ford, W.F. 1982, "Numerical Comparisons of Nonlinear Convergence Accelerators," *Mathematics of Computation*, vol. 38, pp. 481–499.[5]
- Goodwin, E.T. (ed.) 1961, *Modern Computing Methods*, 2nd ed. (New York: Philosophical Library), Chapter 13 [van Wijngaarden's transformations].[6]